AS · Cambridge (CIE) · Physics

1 hour 8 questions

Structured Questions

SI Units

SI Units / Homogeneity of Physical Equations & Powers of Ten

Easy (3 questions)	/35
Medium (3 questions)	/25
Hard (2 questions)	/25
Total Marks	**/85**

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Easy Questions

1 (a) Table 1.1 shows some SI base quantities and their corresponding unit.

Table 1.1

quantity	SI base unit
mass	kg
time	
length	

(i) Complete Table 1.1 to show the SI base unit of each quantity.

[2]

(ii) State two other SI base quantities and their corresponding units.

[2]

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..... **(4 marks)**

(b) Newton's second law of motion describes the relationship between force *F* mass *m* and acceleration *a*:

$$F = ma$$

The derived unit for force is the newton (N).

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Use Newton's second law to write 1 N in SI units.

.....

..... **(2 marks)**

(c) The kinetic energy equation describes the relationship between energy *E*, mass, *m*, and velocity, *v*.

$$E_k = \frac{1}{2}mv^2$$

The derived unit for energy is the joule (J).

Use the equation linking kinetic energy and speed to write 1 J in SI base units.

.....

..... **(2 marks)**

(d) The derived unit for power *P* is the watt W.

Given that power is the energy transferred *E* per unit time *t*, deduce the SI base units for power

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..... **(2 marks)**

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2 (a) In the following list, underline all the units that are SI base units

kilogram newton metre coulomb volt ampere

.....

.....

..... **(3 marks)**

(b) Complete Table 1.1 to show the name and symbol of each prefix and the corresponding power of ten.

Table 1.1

prefix	power of ten
tera (T)	
	10^{9}
	10^{3}
centi (c)	10^{-2}
milli (m)	
micro (μ)	
	10^{-9}
pico (p)	10^{-12}

(3 marks)

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(c) Convert the following measurements into standard form and to the specified number of significant figures.

i) 0.71 GJ to 3 s. f.

[1]

ii) 24.5 kW to 4 s .f.

[1]

iii) 0.003823 mm to 1 s. f.

[1]

iv) 644.9 pN to 2 s.f.

[1]

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..... **(4 marks)**

(d) Convert the following from their standard form into measurements with the most suitable prefixes

9.3×10^{6} J = ... J

7.4×10^{3} W = ... W

1.0×10^{-2} m = ... m

5.3×10^{-9} C = ... C

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(4 marks)

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3 (a) Scientists around the world use the same internationally agreed system of units. These are called the fundamental SI base units and are listed in Fig. 1.1.

Quantity	Name	Symbol
length		
	kilogram	
		s
	ampere	
temperature		K

Fig. 1.1

Complete the missing quantities, names and symbols in Fig. 1.1.

(5 marks)

(b) Use the equation linking kinetic energy and speed to write 1 Joule in SI base units.

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..... **(2 marks)**

(c) Use the equation *Charge* = *Current* × *Time* to write 1 Coulomb in SI base units.

.....

..... **(2 marks)**

(d) Use the equation *Energy* = *Charge* × *Potential Difference* to write 1 volt in SI base units.

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(2 marks)

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Medium Questions

1 (a) Define the following quantities and state their SI base units

(i) velocity

[2]

(ii) acceleration

[2]

(iii) density

[2]

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..... **(6 marks)**

(b) The pressure P due to a liquid of density ρ is related to the depth h by the expression

$$P = \rho g h$$

where g is the acceleration of free fall.

Use this expression to determine the derived units of pressure P.

.....

..... **(2 marks)**

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(c) The speed v of a sound wave through a gas of pressure P and density ρ is given by the equation

$$v = \sqrt{\frac{\gamma P}{\rho}}$$

where γ is a constant.

Show that the constant γ has no unit.

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..... **(3 marks)**

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2 (a) Derive the SI base unit of force.

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(2 marks)

(b) As a spherical object of radius r moves through the air at speed v, it experiences a force F due to air resistance.

The resistive force F is given by

$$F = krv$$

Where k is a constant.

Derive the SI base unit of the constant k.

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(2 marks)

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3 (a) Fig 1.2 shows a turbine that is used to generate electrical power *P* from wind travelling with velocity *v*.

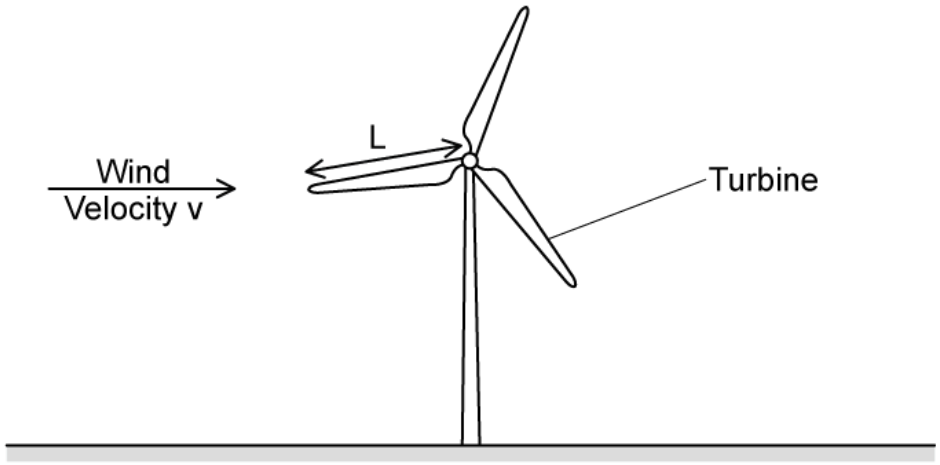


Fig. 1.2

The useful power output *P* of the turbine is given by

$$P = C_P L^2 \rho v^3$$

Where *L* is the length of each blade, ρ is the density of air, *v* is the wind velocity and C_P is a constant.

Show that the constant C_P has no units.

(5 marks)

(b) A local council is looking into building wind turbines to supply power to 2500 households in a village. The average energy consumption of a household in the village is 2800 kWh a year.

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Fig. 1.3 shows an image from a catalogue detailing the different sizes and power outputs of available wind turbines.

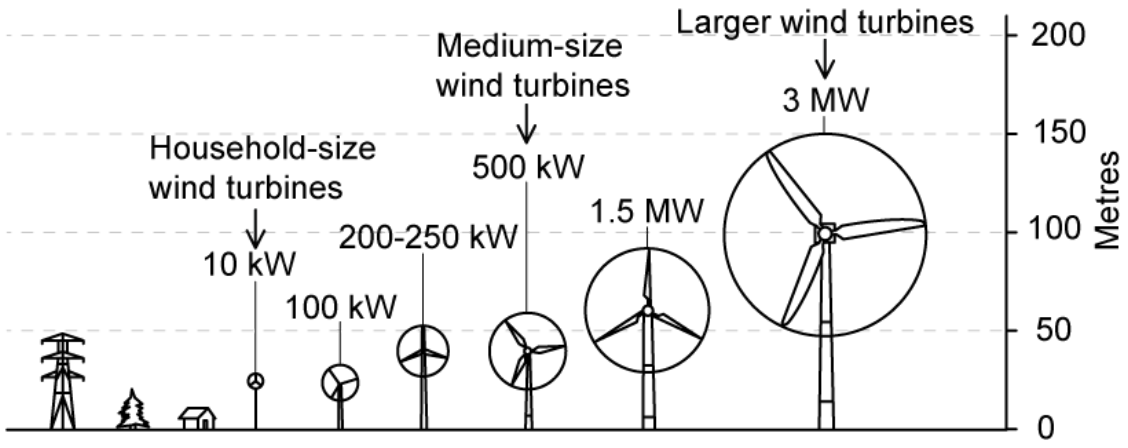


Fig. 1.3

They decide to build 8 identical wind turbines to supply the entire village with electricity.

(i) Determine which of the wind turbines in Fig. 1.3 the council decided to build.

[3]

(ii) The chosen turbine delivers the power specified at a C_P value of 0.45. This value can only be achieved above certain wind velocities.

Estimate the average wind velocity needed to power the village for a year. The density of air is 1.3 kg m^{-3}.

[2]

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..... **(5 marks)**

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Hard Questions

1 (a) Convert the following measurements

(i) 42 km^3 into mm^3

[2]

(ii) 25 light-years into nm

[2]

(iii) 0.05 kWh into PeV

[2]

(6 marks)

(b) The frequency f of a stationary wave on a string is given by

$$f^k = \frac{T}{4mL^2}$$

where L is the length of the string, T is the tension in the string, m is the mass per unit length of the string and k is a dimensionless constant.

By considering the homogeneity of the equation, determine the value of k.

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(3 marks)

(c) The current I through a semiconductor diode given by

$$I = I_0 e^{-\frac{qV}{nT}}$$

where I_0 is the initial current, q is the charge of an electron, V is the applied potential difference, T is the temperature and n is a constant.

(i) By considering the homogeneity of the equation, determine the units of n.

[3]

(ii) Use the datasheet to determine the name of this constant.

[1]

(4 marks)

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2 (a) The diameter of a proton is approximately 1 fm.

Estimate the time it takes light to cross the nucleus of a hydrogen atom.

$1 \text{ fm} = 1 \times 10^{-15} \text{ m}$

(2 marks)

(b) A foil made from gold has a thickness of 0.132 μm. A gold atom has a radius of 146 pm.

(i) Estimate the number of layers of atoms in the gold foil to the nearest hundred.

[1]

(ii) The radius of a gold nucleus is about 20 000 times smaller than its atomic radius. The cross-sectional area of a nucleus is commonly measured using the unit *barn*, which is represented by the symbol *b*.

Estimate the cross-sectional area of a gold nucleus in barns (b).

$1 \text{ b} = 100 \text{ fm}^2$

[3]

(4 marks)

(c) The atomic mass unit *u* is a common unit used in nuclear physics. 1*u* is equivalent to 931.5 MeV c^{-2}

Show that 1*u* is approximately equal to the mass of a proton in kg.

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(3 marks)

(d) An acre is a unit of land measurement approximately equal to 4050 m^2, which is roughly equivalent to the size of a football field.

An Astronomical Unit, or 1 AU, is defined as the average distance between the Earth and the Sun and is approximately equal to 150 million km.

Estimate the number of football fields that could fit lengthwise between the Earth and the Sun.

(3 marks)

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